Box Plot of Salary Data.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Han...

```python
import pandas as pd
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("employee.csv")

# Remove rows where Salary is missing
df = df.dropna(subset=["Salary"])

# Find outliers
Q1 = df["Salary"].quantile(0.25)
Q3 = df["Salary"].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Salary"] < lower) | (df["Salary"] > upper)]

# Print output BEFORE showing graph
print("\nOutliers:")
print(outliers)

# Draw box plot
plt.figure(figsize=(6,4))
plt.boxplot(df["Salary"])

plt.title("Salary Box Plot")
plt.ylabel("Salary")

# Show graph LAST
plt.show()
```

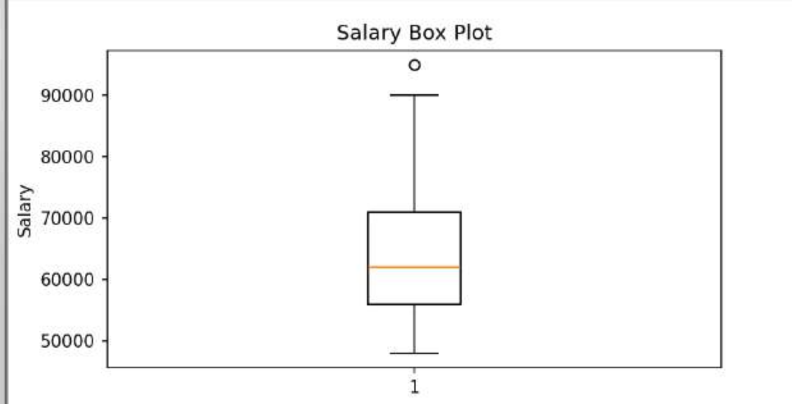
Ln: 33 Col: 10

IDLE Shell 3.12.10

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Dat
a Summarization/Box Plot of Salary Data.py

Outliers:
   EmployeeID        Name Department  ...   Salary  YearsExperience  Joi
ningDate
6        107  Vikram Rao    Finance  ...  95000.0             18.0   20
07-07-19

[1 rows x 7 columns]
```

Ln: 5 Col: 0

Figure 1



Detect Outliers using IQR Method.py - C:/Users/irsha/Documents/DSA0405 programs/...

File Edit Format Run Options Window Help

```python
import pandas as pd

# Read CSV file
df = pd.read_csv("student_marks.csv")

# Calculate Quartiles
Q1 = df["Marks"].quantile(0.25)
Q3 = df["Marks"].quantile(0.75)
IQR = Q3 - Q1

# Calculate Limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

# Detect Outliers
outliers = df[(df["Marks"] < lower_limit) | (df["Marks"] > upper_limit)]

print("Student Marks Dataset")
print(df)

print("\nQ1 =", Q1)
print("Q3 =", Q3)
print("IQR =", IQR)

print("\nLower Limit =", lower_limit)
print("Upper Limit =", upper_limit)

if outliers.empty:
    print("\nNo Outliers Found.")
else:
    print("\nOutliers Detected:")
    print(outliers)

print("\nProgram Completed.")
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Dat
a Summarization/Detect Outliers using IQR Method.py
Student Marks Dataset
    Student_ID     Name  Marks
0          101    Alice     65
1          102      Bob     70
2          103  Charlie     72
3          104    David     68
4          105      Eva     75
5          106    Frank     80
6          107    Grace     77
7          108    Henry     74
8          109      Ivy     69
9          110     Jack     73
10         111     Kate     78
11         112     Liam     82
12         113      Mia     71
13         114     Noah     76
14         115   Olivia     98

Q1 = 70.5
Q3 = 77.5
IQR = 7.0

Lower Limit = 60.0
Upper Limit = 88.0

Outliers Detected:
    Student_ID    Name  Marks
14         115  Olivia     98

Program Completed.
>>>
```

Employee dataset.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on...

File Edit Format Run Options Window Help

```python
import pandas as pd

# Load dataset
df = pd.read_csv("employee.csv")

print("========== EMPLOYEE DATASET SUMMARY ==========")

# Number of rows
print("\nNumber of Rows :", df.shape[0])

# Number of columns
print("Number of Columns :", df.shape[1])

# Data types
print("\nData Types")
print(df.dtypes)

# Missing values
print("\nMissing Values")
print(df.isnull().sum())

# Statistical summary
print("\nStatistical Summary")
print(df.describe(include='all'))
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Dat
a Summarization/Employee dataset.py
========== EMPLOYEE DATASET SUMMARY ==========

Number of Rows : 15
Number of Columns : 7

Data Types
EmployeeID           int64
Name                   str
Department             str
Age                float64
Salary             float64
YearsExperience    float64
JoiningDate            str
dtype: object

Missing Values
EmployeeID         0
Name               0
Department         0
Age                2
Salary             2
YearsExperience    1
JoiningDate        0
dtype: int64

Statistical Summary
        EmployeeID          Name  ... YearsExperience  JoiningDate
count    15.000000            15  ...       14.000000           15
unique         NaN            15  ...             NaN           15
top            NaN  Aarav Sharma  ...             NaN   2020-03-15
freq           NaN             1  ...             NaN            1
mean    108.000000           NaN  ...        8.428571          NaN
std       4.472136           NaN  ...        5.487248          NaN
min     101.000000           NaN  ...        1.000000          NaN
25%     104.500000           NaN  ...        4.250000          NaN
50%     108.000000           NaN  ...        7.500000          NaN
75%     111.500000           NaN  ...       13.000000          NaN
max     115.000000           NaN  ...       18.000000          NaN

[11 rows x 7 columns]
>>>
```

Exploratory Data Analysis using Iris Dataset.py - C:\Users\irsha\Documents\DSA0405 p...

File Edit Format Run Options Window Help

```python
import pandas as pd
import matplotlib.pyplot as plt

# Read Titanic Dataset
df = pd.read_csv("Titanic-Dataset.csv")

# Dataset Information
print("========== DATASET INFORMATION ==========")
df.info()

# Missing Values
print("\n========== MISSING VALUES ==========")
print(df.isnull().sum())

# Descriptive Statistics
print("\n========== DESCRIPTIVE STATISTICS ==========")
print(df.describe())

# Remove missing Age values
age = df["Age"].dropna()

# Detect Outliers using IQR
Q1 = age.quantile(0.25)
Q3 = age.quantile(0.75)
IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

outliers = df[(df["Age"] < lower_limit) | (df["Age"] > upper_limit)]

print("\n========== OUTLIERS ==========")
print(outliers)

# Remove Outliers
cleaned_df = df[(df["Age"] >= lower_limit) & (df["Age"] <= upper_limit)]

print("\n========== CLEANED DATASET ==========")
print(cleaned_df.head())

# Save Cleaned Dataset
cleaned_df.to_csv("Titanic_Cleaned.csv", index=False)

print("\nCleaned dataset saved as Titanic_Cleaned.csv")
print("Program Completed.")

# Create both graphs in one window
plt.figure(figsize=(12,5))
```

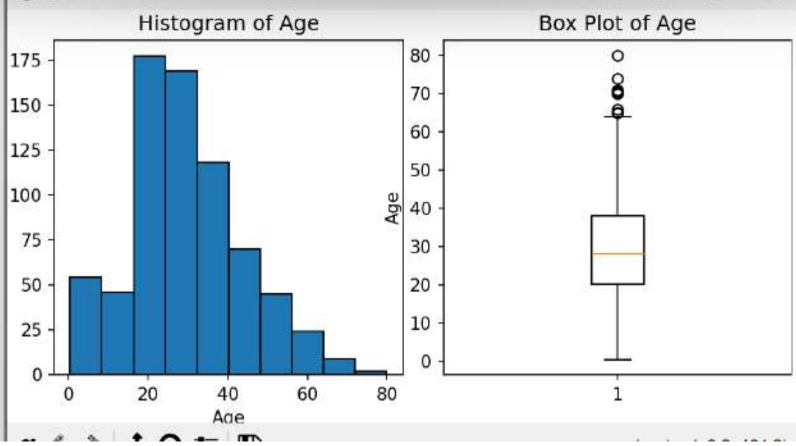
IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
25%      223.500000    0.000000    2.000000  ...    0.000000    0.000000
  7.910400
50%      446.000000    0.000000    3.000000  ...    0.000000    0.000000
 14.454200
75%      668.500000    1.000000    3.000000  ...    1.000000    0.000000
 31.000000
max      891.000000    1.000000    3.000000  ...    8.000000    6.000000
512.329200

[8 rows x 7 columns]

========== OUTLIERS ==========
     PassengerId  Survived  Pclass  ...     Fare Cabin  Embarked
33            34         0       2  ...  10.5000   NaN         S
54            55         0       1  ...  61.9792   B30         C
96            97         0       1  ...  34.6542    A5         C
116          117         0       3  ...   7.7500   NaN         Q
280          281         0       3  ...   7.7500   NaN         Q
456          457         0       1  ...  26.5500   E38         S
493          494         0       1  ...  49.5042   NaN         C
630          631         1       1  ...  30.0000   A23         S
672          673         0       2  ...  10.5000   NaN         S
```

Ln: 82 Col: 0

Figure 1



Frequency Distribution of Students' Grades.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on

File Edit Format Run Options Window Help

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("students.csv")

# Frequency distribution
grade_count = df["Grade"].value_counts().sort_index()

print(grade_count)

# Bar Chart
grade_count.plot(kind="bar")

plt.title("Frequency Distribution of Grades")
plt.xlabel("Grades")
plt.ylabel("Frequency")
plt.show()
```

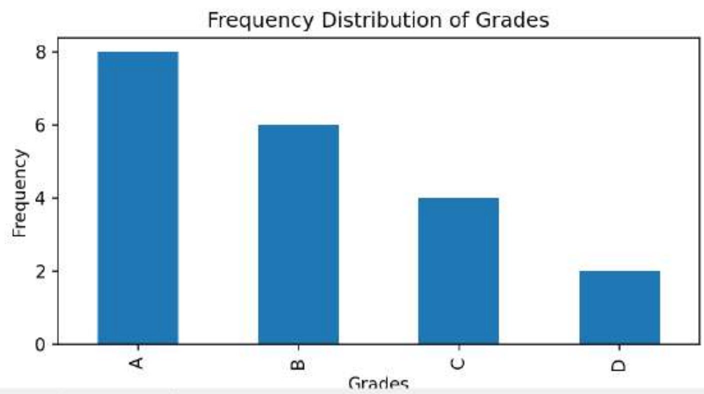
IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Data Summarization/Frequency Distribution of Students' Grades.py
Grade
A    8
B    6
C    4
D    2
Name: count, dtype: int64
```

Ln: 5 Col: 0

Figure 1



Histogram of House Prices.py - C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Data Sum

File Edit Format Run Options Window Help

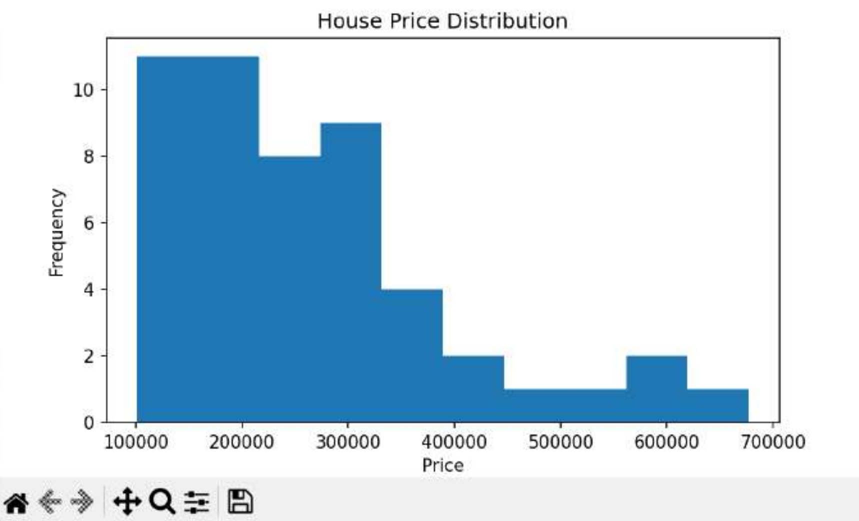
```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("house.csv")

# Identify skewness
skew = df["Price"].skew()

print("Skewness =", round(skew, 2))

if abs(skew) < 0.5:
    print("Approximately Normally Distributed")
elif skew > 0:
    print("Positively Skewed")
else:
    print("Negatively Skewed")

# Histogram
plt.figure(figsize=(6,4))
plt.hist(df["Price"], bins=10)

plt.title("House Price Distribution")
plt.xlabel("Price")
plt.ylabel("Frequency")

plt.show()
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64 bit
 (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Data Summar
ization/Histogram of House Prices.py
Skewness = 1.27
Positively Skewed
```

Figure 1



Histograms and Box Plots Before & After Removing Outliers.py - C:/Users/irsha/Docum...

File Edit Format Run Options Window Help

```python
IQR = Q3 - Q1

# Calculate Limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

# Remove Outliers
cleaned_df = df[(df["Sales"] >= lower_limit) & (df["Sales"] <= upper_limit)]

print("Original Dataset")
print(df)

print("\nLower Limit =", lower_limit)
print("Upper Limit =", upper_limit)

print("\nCleaned Dataset")
print(cleaned_df)

# Histogram Before Removing Outliers
plt.figure(figsize=(8,5))
plt.hist(df["Sales"], bins=10, edgecolor="black")
plt.title("Histogram Before Removing Outliers")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

# Box Plot Before Removing Outliers
plt.figure(figsize=(6,5))
plt.boxplot(df["Sales"], vert=True)
plt.title("Box Plot Before Removing Outliers")
plt.ylabel("Sales")
plt.show()

# Histogram After Removing Outliers
plt.figure(figsize=(8,5))
plt.hist(cleaned_df["Sales"], bins=10, edgecolor="black")
plt.title("Histogram After Removing Outliers")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

# Box Plot After Removing Outliers
plt.figure(figsize=(6,5))
plt.boxplot(cleaned_df["Sales"], vert=True)
plt.title("Box Plot After Removing Outliers")
plt.ylabel("Sales")
plt.show()

print("\nProgram Completed.")
```

Ln: 16 Col: 0

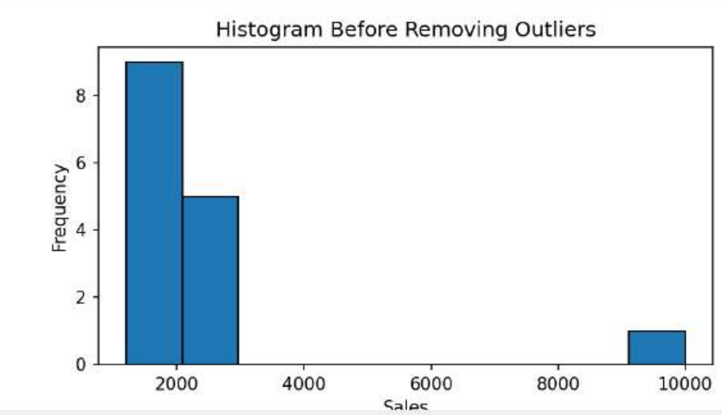
IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Dat
a Summarization/Histograms and Box Plots Before & After Removing Outlier
s.py
Original Dataset
    Product_ID  Sales
0          101   1200
1          102   1350
2          103   1400
3          104   1500
4          105   1600
5          106   1700
6          107   1800
7          108   1900
8          109   2000
9          110   2100
10         111   2200
11         112   2300
12         113   2400
13         114   2500
14         115  10000

Lower Limit = 500.0
```

Ln: 5 Col: 0

Figure 1



Remove Outliers using IQR Method.py - C:/Users/irsha/Documents/DSA0405 program...

File Edit Format Run Options Window Help

```python
import pandas as pd

# Read CSV file
df = pd.read_csv("sales_data.csv")

# Calculate Quartiles
Q1 = df["Sales"].quantile(0.25)
Q3 = df["Sales"].quantile(0.75)
IQR = Q3 - Q1

# Calculate Limits
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

# Detect Outliers
outliers = df[(df["Sales"] < lower_limit) | (df["Sales"] > upper_limit)]

print("Original Sales Dataset")
print(df)

print("\nLower Limit =", lower_limit)
print("Upper Limit =", upper_limit)

print("\nOutliers:")
print(outliers)

# Remove Outliers
cleaned_df = df[(df["Sales"] >= lower_limit) & (df["Sales"] <= upper_limit)]

print("\nCleaned Dataset")
print(cleaned_df)

print("\nProgram Completed.")
```

IDLE Shell 3.12.10

File Edit Shell Debug Options Window Help

```
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.194
3 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/Unit 3/Hands on Dat
a Summarization/Remove Outliers using IQR Method.py
Original Sales Dataset
    Product_ID  Sales
0          101   1200
1          102   1350
2          103   1400
3          104   1500
4          105   1600
5          106   1700
6          107   1800
7          108   1900
8          109   2000
9          110   2100
10         111   2200
11         112   2300
12         113   2400
13         114   2500
14         115  10000

Lower Limit = 500.0
Upper Limit = 3300.0

Outliers:
    Product_ID  Sales
14         115  10000

Cleaned Dataset
    Product_ID  Sales
0          101   1200
1          102   1350
2          103   1400
3          104   1500
4          105   1600
5          106   1700
6          107   1800
7          108   1900
8          109   2000
9          110   2100
10         111   2200
11         112   2300
12         113   2400
13         114   2500

Program Completed.
>>>
```